

QA DISPOSITION REPORT burn-in parametric screening

Component	LOT192-C0144	Lot	LOT192	Board	LOT192-B02
Socket	r2 c0	Model version	dataset-v1.1 / ModuleA-d13e1ef / ModuleB-2628c09261	2023-09-26 19:01	

DECISION: WATCH

LAYER 1 — deterministic evidence (every value cites its comparison basis)

Absolute limit	PASS (51.54 mA vs 65.0 mA max, supply_current_ma)
Lot-relative level	-3.6 robust sigma vs lot median on prop_delay_ns (median 8.19 ns, sigma 0.31 ns, n=499, MAD estimator; AEC-Q001 IQR/1.35 gives 0.32 ns)
Drift 0h to 24h	-0.333 mV on vth_shift_mv; lot median drift +0.926 mV, part is -1.6 robust sigma on the lot's drift distribution
Drift rate	-0.0139 mV/h (lot median 0.0386 mV/h, -0.4x)
Max interval jump	+0.6 robust sigma on prop_delay_ns; largest interval change is 2.9x the median of the others (spread -> progressive drift)
Predicted 168h	6.912 ns (prop_delay_ns, from 0h and 24h only)
95% upper bound	7.118 ns -> within lot-derived safe limit 10.000 ns
Joint anomaly	Mahalanobis D2 = 15.1 (good-part 99th pct = 18.1); largest contribution prop_delay_ns at 50% of the total
Leakage/delay sign	leakage +3.2 sigma, delay -3.6 sigma -- consistent with the lot's process locus
Spatial check	no neighbour clustering on board LOT192-B02 (0/64 flagged), component-specific
Data quality	all checkpoints valid
Primary reason	no rule fired
Action	Release, but tag the component ID for trend monitoring at the next screen. No rework.

LAYER 2 — model attribution

detector	normalised strength	fired	feature (Module B)	SHAP (mV)
L3b_T2	1.28	YES	v24	-10.059
L1_dynamic_MAD	1.19	YES	z:d1	+0.367
L3c_kNN	1.04	YES	d1	-0.336
L3a_MCD	0.84	no	z:v0	-0.201
L4d_AutoEnc	0.68	no	lotmed_t0	-0.093

Detector strength is distance past that detector's own operating threshold (1.0 = exactly at it). SHAP values are exact TreeSHAP from the LightGBM drift model, in target units. **Caveat:** v0 and v24 are 0.99 correlated, so SHAP credit between them is not identifiable — a Huber model of equal accuracy attributes it the other way round. Treat Layer 2 as supporting evidence; Layer 1 is the auditable record.

COUNTERFACTUAL

No single-parameter change clears the rules; the evidence is distributed across several measurements.

LAYER 3 — failure-mechanism HYPOTHESIS (for the FA engineer; not a diagnosis)

No mechanism signature matched; no hypothesis offered.

SIGN-OFF

Inspector		Date	
Disposition		Signature	
Notes			

Decision support only. Final rejection remains under approved engineering rules and QA control; this system detects, ranks, explains and forecasts. The model never overrides a hard engineering limit in either direction.